

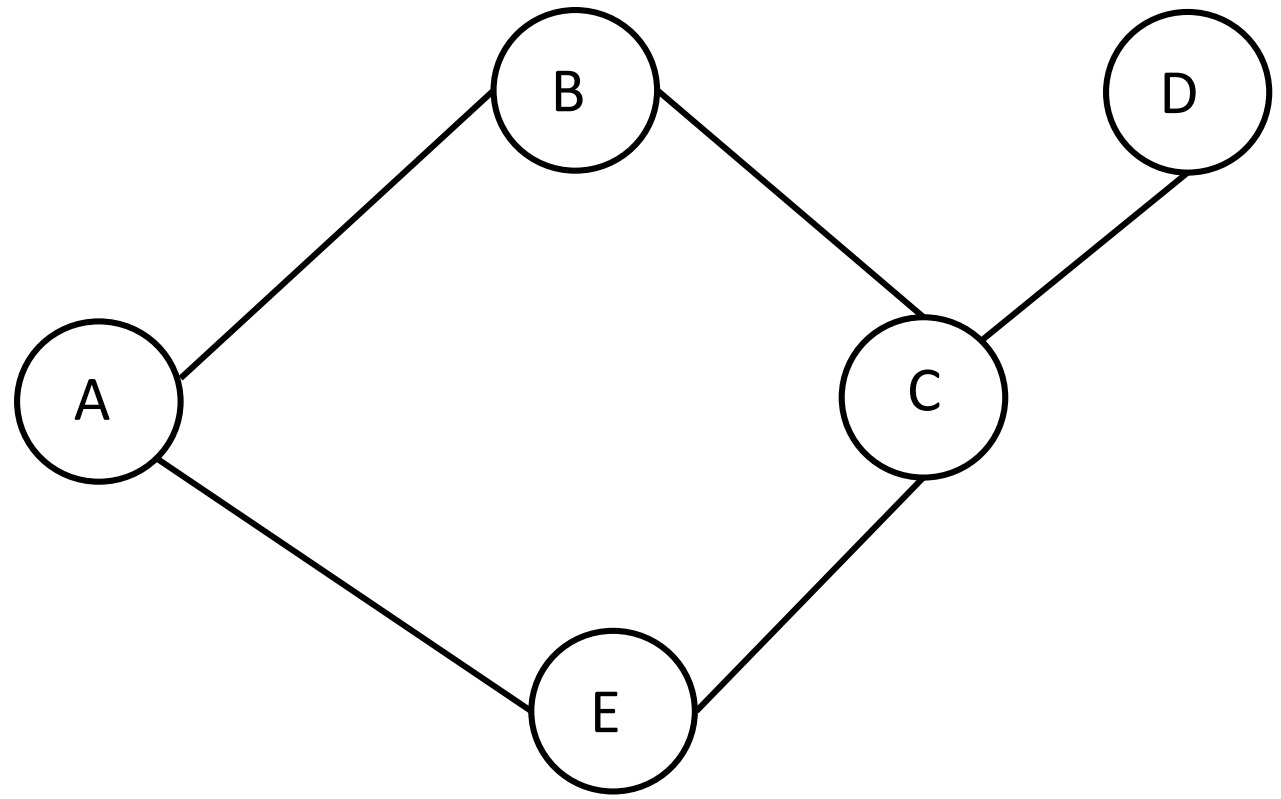
All Pair Shortest Path

Floyd Warshall Algorithm

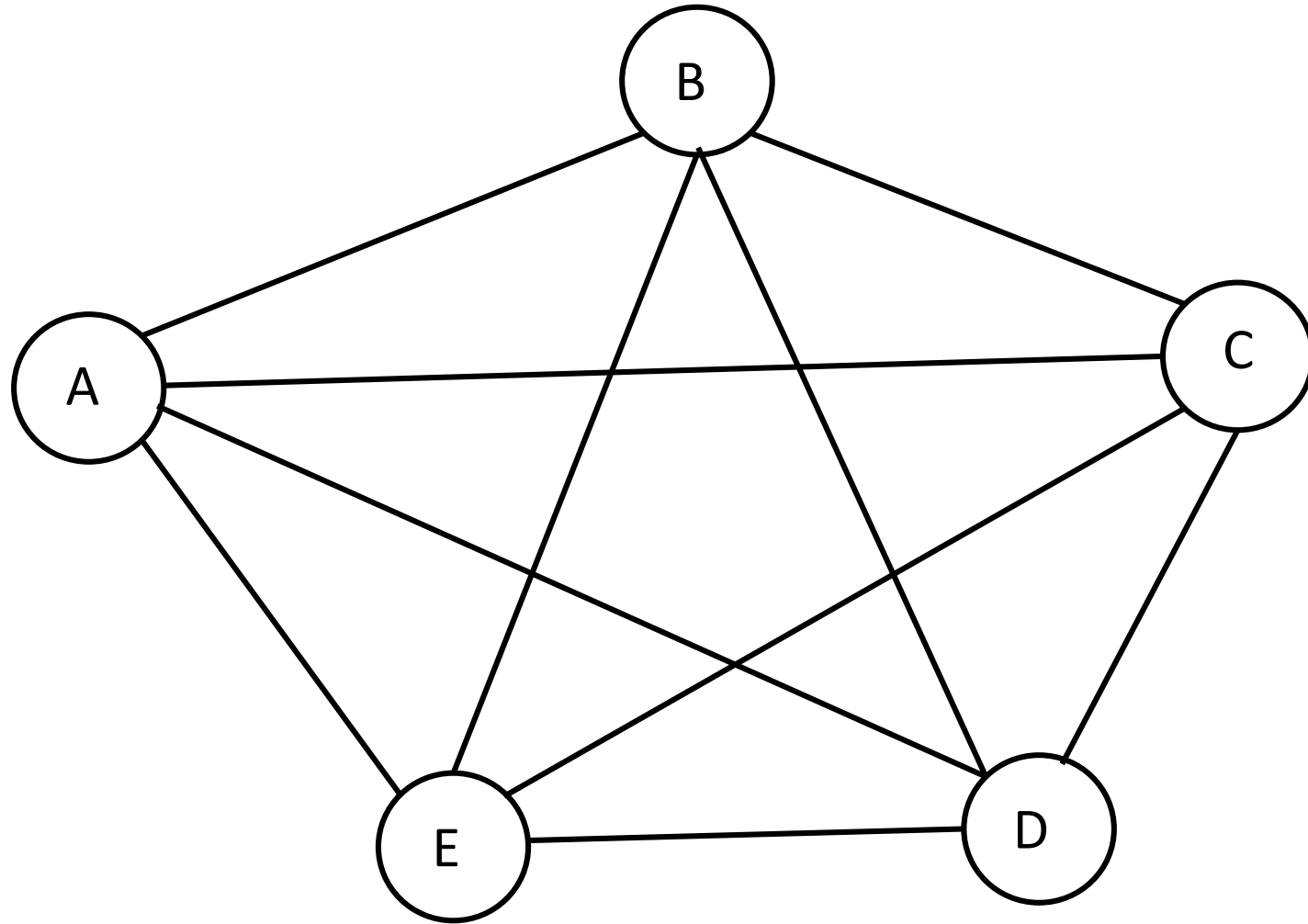
All Pair Shortest Path

The **Floyd-Warshall Algorithm** is a dynamic programming algorithm used to find the shortest paths between all pairs of vertices in a weighted graph, unlike [Dijkstra](#) and [Bellman-Ford](#) which are single source shortest path algorithms.. It works for both directed and undirected graphs and can handle graphs with both **positive** and **negative** weight edges.

In Dijkstra



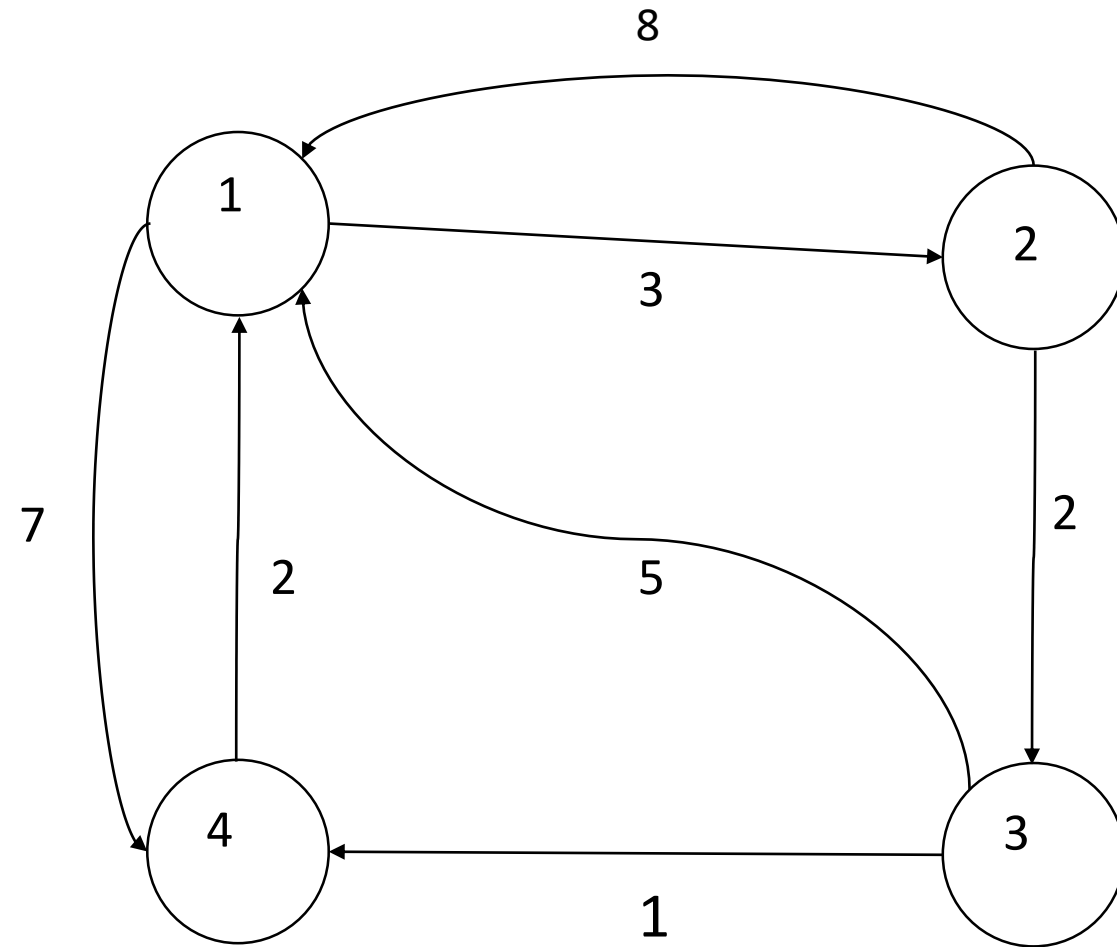
In Floyd Warshall



Algorithm

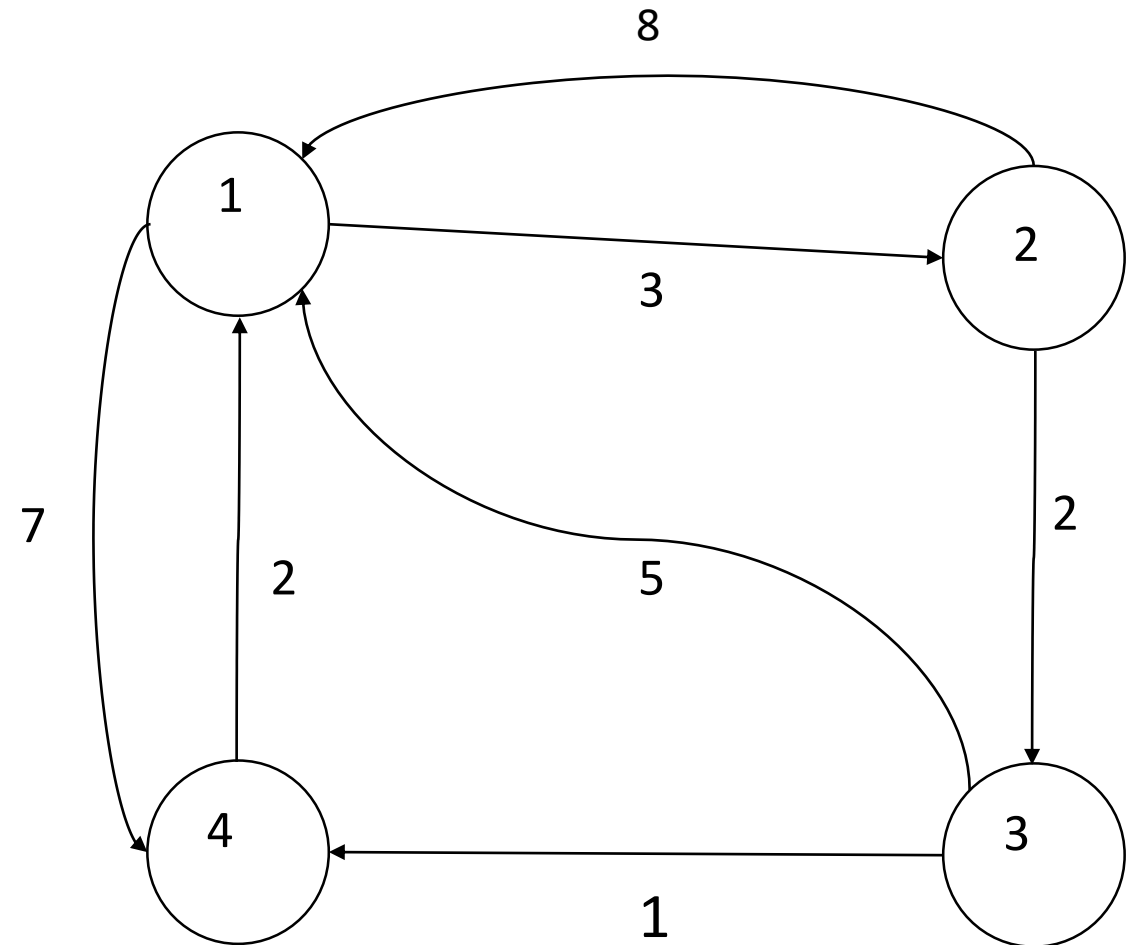
1. Create a distance matrix $\text{dist}[][]$, where $\text{dist}[i][j]$ holds the weight of the edge between node i and j . If no edge exists set it to infinity. Set $\text{dist}[i][i]$ to 0.
2. For each node k (acting as a intermediate node),
For each pair of node (i, j)
Update **$\text{dist}[i][j] = \min(\text{dist}[i][j], \text{dist}[i][k] + \text{dist}[k][j])$** .
3. After processing all nodes, **$\text{dist}[i][j]$** gives the shortest distance from i to j .

Problem:



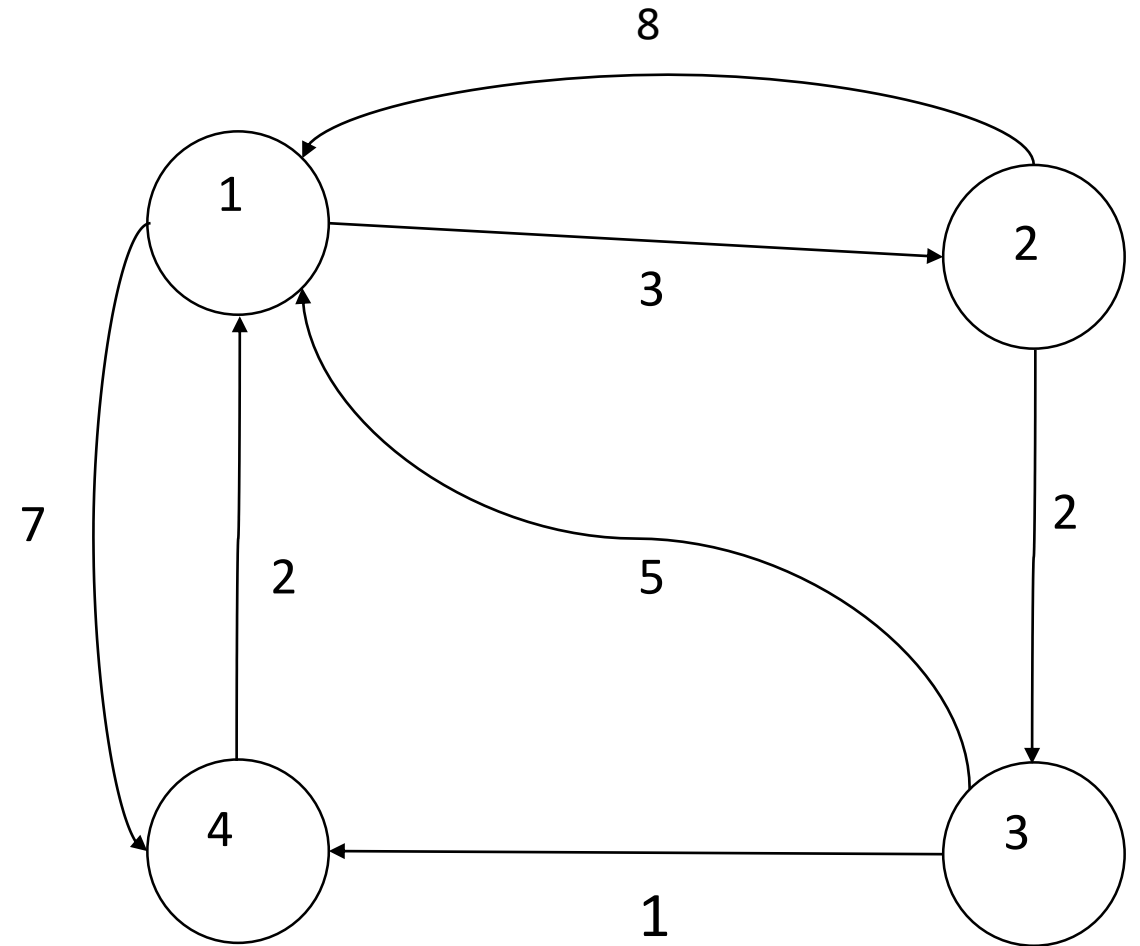
Problem:

$$H^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & \infty \\ 5 & \infty & 0 & 1 \\ 2 & \infty & \infty & 0 \end{pmatrix} \end{matrix}$$



Problem:

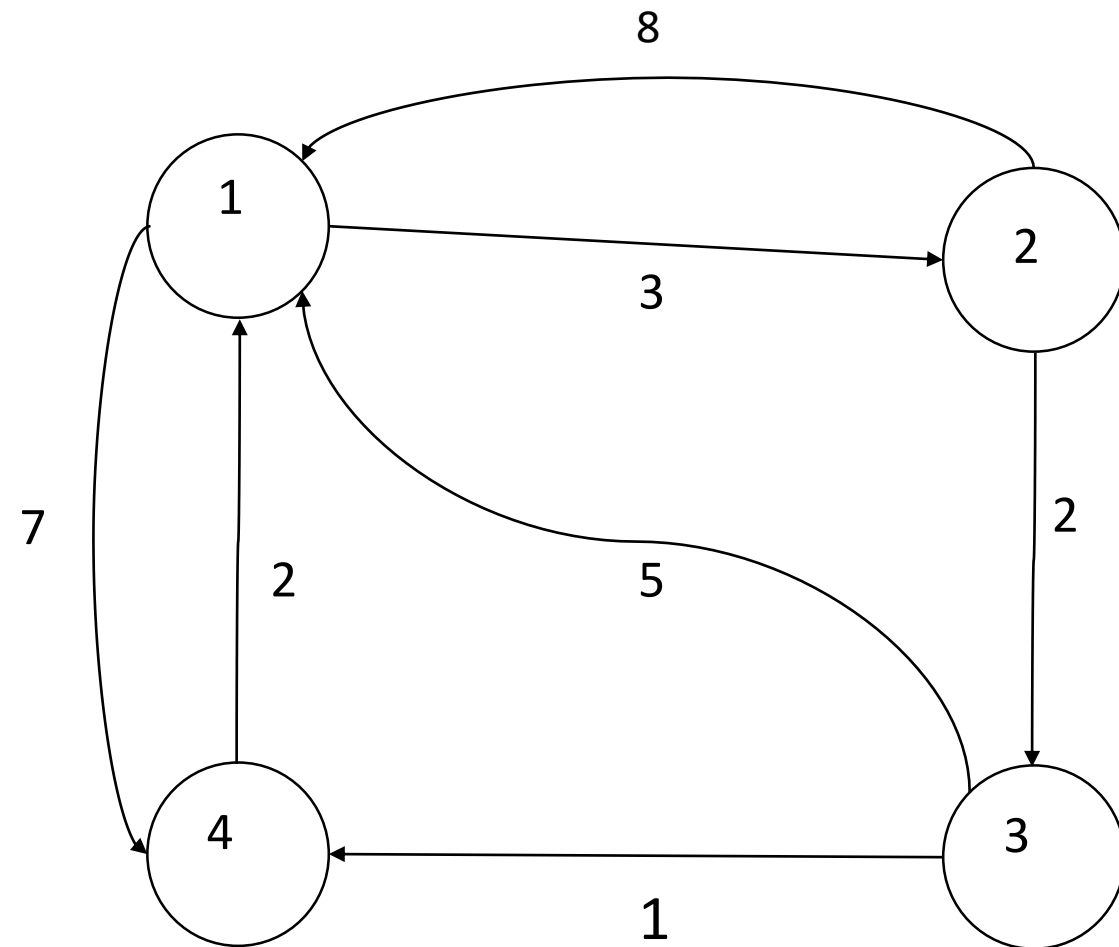
$$H1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & 15 \\ 5 & 8 & 0 & 1 \\ 2 & 5 & \infty & 0 \end{pmatrix} \end{matrix}$$



Problem:

H2 =

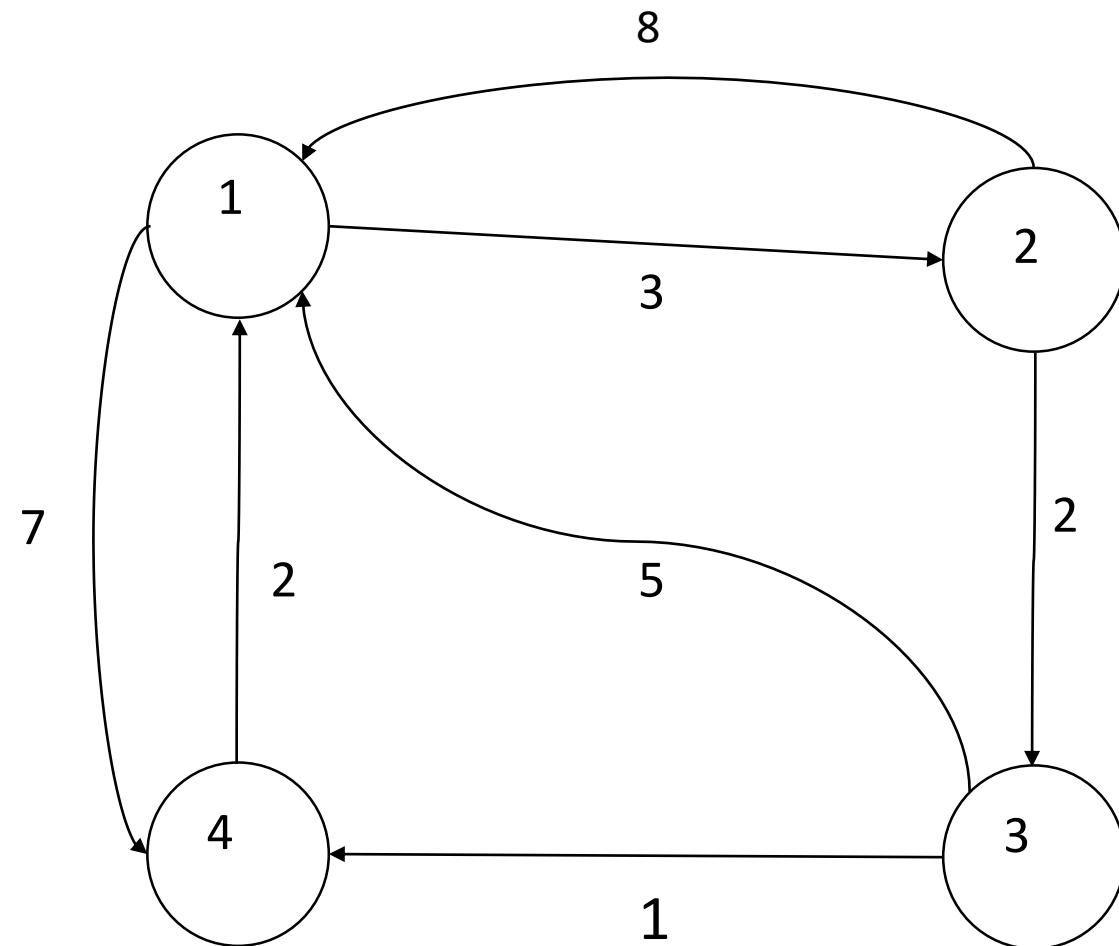
	1	2	3	4
1	0	3	5	7
2	8	0	2	15
3	5	8	0	1
4	2	5	7	0



Problem:

H3 =

	1	2	3	4
1	0	3	5	6
2	7	0	2	3
3	5	8	0	1
4	2	5	7	0



Problem:

$$H_4 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0 & 3 & 5 & 6 \\ 5 & 0 & 2 & 3 \\ 3 & 6 & 0 & 1 \\ 2 & 5 & 7 & 0 \end{pmatrix} \end{matrix}$$

